This wood ant measures 4 million nanometers in length.

Nanoscale

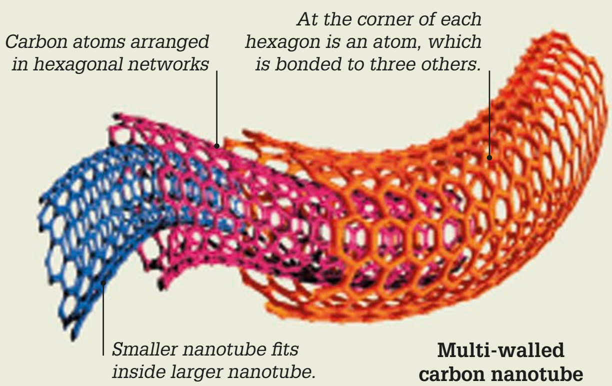
The nanoscale is usually thought of in the range of 1 to 100 nanometers (nm). Objects within the nanoscale include viruses, the width of DNA strands, and many molecules, while a single hydrogen atom is approximately 0.1 nm across.

Nanotechnology

Nanotechnology is the science and engineering of working at a scale utterly invisible to the human eye. A nanometer (nm) is a billionth of a meter—so there are a billion nanometers in one meter. To convey just how tiny nanometers are, this book page is around 100,000 nm thick. Working at such a scale is still in its infancy and mostly at the research stage, but future advances could have enormous impacts on materials science, medicine, robotics, and computing.

“It is a staggeringly small world that is below.”

Richard Feynman, US physicist in his lecture *There’s Plenty of Room at the Bottom*, 1959



Multi-walled carbon nanotube

Nanomaterials

Materials constructed at the nanoscale can have valuable properties such as great strength, lightness, the ability to repel water or bacteria, or to conduct electricity or heat extremely well. Carbon nanotubes (cylinders of carbon atoms), for example, are much lighter than steel, but more than 100 times stronger.

Nanoparticles

Particles at the nanoscale have been added to substances to alter their properties in some way. Titanium dioxide nanoparticles, for example, help to block harmful UV (ultraviolet) rays, but do not reflect visible light. They are used in transparent sunscreen lotions.

Silver nanoparticles

These nanoparticles can repel and kill bacteria, making them useful in antimicrobial wound dressings (left). They are also used in some sports shoes to kill bacteria that can cause odors.

Titanium dioxide nanoparticles

This air-purifying unit contains nanoparticles of titanium dioxide. When sprayed, they react with UV radiation and water in the air, to break down pollutants.

Key events

1959

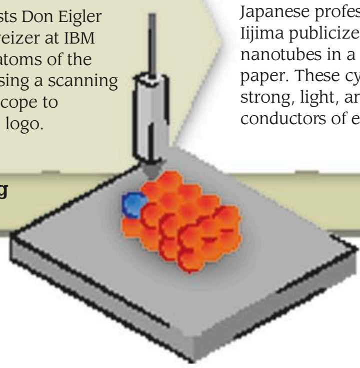
American physicist Richard Feynman gave a pioneering early lecture entitled *There’s Plenty of Room at the Bottom*. It focused on engineering and technology that could “arrange the atoms the way we want.”

1974

The term nanotechnology was used for the first time by Tokyo Science University’s Professor Norio Taniguchi to describe working with materials at an atomic scale. It was popularized in the US by American engineer Eric Drexler.

1989

American scientists Don Eigler and Erhard Schweizer at IBM manipulated 35 atoms of the element xenon using a scanning tunneling microscope to spell out the IBM logo.



Scanning tunneling microscope moving atoms

1991

Japanese professor Sumio Iijima publicized carbon nanotubes in a scientific paper. These cylinders are strong, light, and excellent conductors of electricity.